UNIT 1 PRE-TEST · FORM B

The Real Number System

Answer each question, then check your work.

Name _____ Date _____ Period _____

SCORE 3.0 Basic Understanding

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

1. Write as a decimal: $5/8$

= _____

e.g. 0.5

2. Write as a fraction: 0.75

= _____

simplest form, e.g. 3/4

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

3. Classify: $\sqrt{169}$

Classification: _____

type R or IR

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

4. Write as a fraction: $0.\overline{3}$ (the 3 repeats)

= _____

simplest form

5. Write as a decimal: 2⅓ (2 and 1/3)

= _____

use ... for the repeating part

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{50}$

Classification: _____

type R or IR

Simplify each expression first, then classify the result.

7. Classify: $\sqrt{2} \times \sqrt{8}$

Classification: _____

simplify, then R or IR

8. Classify: $\sqrt{75} \div \sqrt{3}$

Classification: _____

simplify, then R or IR

1.3 Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

9. Estimate: $\sqrt{28}$



10. Estimate: $\sqrt{60}$



SCORE 5.0 Advanced Content

1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

11. Classify: $2 + \sqrt{16}$

Classification: _____

simplify, then R or IR

12. Classify: $5 + \sqrt{7}$

Classification: _____

type R or IR

1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

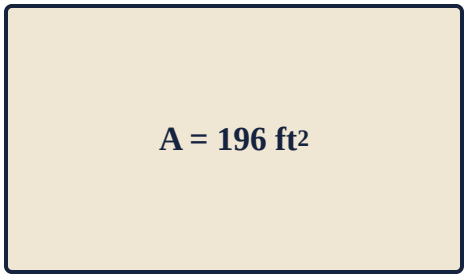
13. Set: { 0.8, $\sqrt{3}$, 1.2, 7/5 }



Order: _____

1.5 Area

A square garden has an area of 196 ft². Find the side length and the perimeter.



14. Find the side length and perimeter of the square garden.

Side length = _____ ft

Perimeter = _____ ft